CHAPTER 1

SEXUAL REPRODUCTION IN FLOWERING PLANTS

CHAPTER ESSENCE AT A GLANCE:
Microsporogenesis and megasporogenesis culminate in double fertilization, establishing the triploid endosperm and diploid embryo.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Microsporogenesis and Pollen Wall Layers:** Anther is bilobed and dithecous with four microsporangia. Wall layers from outside inward: Epidermis, Endothecium (hygroscopic fibrous bands for dehiscence), Middle layers, and Tapetum (multinucleated nourishing layer with dense cytoplasm). Pollen exine is composed of sporopollenin, the most resistant biological material, punctured by germ pores; intine is pectin-cellulose.

• **Megasporogenesis and 7-Celled Embryo Sac:** Megaspore mother cell (MMC) undergoes meiosis producing four megaspores; three chalazal degenerating or one chalazal functional megaspore undergoes three free-nuclear mitotic divisions to form a monosporic 7-celled, 8-nucleate embryo sac (Polygonum type): 3 antipodals at chalazal pole, 2 synergids with filiform apparatus and 1 egg at micropylar pole, and 1 large central cell with 2 polar nuclei.

• **Outbreeding Mechanisms Against Inbreeding:** To prevent inbreeding depression and promote xenogamy, angiosperms utilize outbreeding adaptations: Dichogamy (non-synchronous pollen release and stigma receptivity: protandry or protogyny), Herkogamy (spatial separation of anther and stigma), Self-incompatibility (genetically controlled pollen inhibition on same flower or plant stigma), and Dicliny (unisexual flowers, e.g., dioecious papaya).

• **Double Fertilization and Seed Ontogeny:** Unique to angiosperms: Syngamy fuses one haploid male gamete (n) with the haploid egg (n) to produce a diploid zygote (2n); Triple Fusion fuses the second male gamete (n) with diploid secondary polar nuclei (2n) to form the triploid Primary Endosperm Nucleus (PEN, 3n). Endosperm develops before the zygote to nourish the embryo. Albuminous seeds retain endosperm (maize, castor); exalbuminous seeds consume endosperm (pea, groundnut). Apomixis produces seeds without fertilization.

ESSENTIAL TAXONOMICAL FRAMEWORKS & PROCESSES

$$\text{Double Fertilization} = \text{Syngamy } (n + n \to 2n \text{ Zygote}) + \text{Triple Fusion } (n + 2n \to 3n \text{ PEN})$$
Angiosperm Dual Fertilization Stoichiometry

$$\text{Mature Embryo Sac} = 3 \text{ Antipodals (Chalaza)} + 2 \text{ Synergids} + 1 \text{ Egg} + 1 \text{ Central Cell (2 Nuclei)}$$
Seven-Celled Eight-Nucleate Monosporic Architecture

$$\text{Anther Wall Stratification} = \text{Epidermis} \to \text{Endothecium} \to \text{Middle Layers} \to \text{Tapetum (Nutritive)}$$
Centripetal Stratification of the Microsporangial Wall

$$\text{Exine Armor} = \text{Sporopollenin} \quad [\text{Impervious to acids, alkalis, enzymes; absent at germ pores}]$$
Pollen Grain Protective Biochemical Architecture

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Geitonogamy is functionally cross-pollination (requires an external pollinating agent) but genetically self-pollination because pollen grains originate from the same plant.

• **Alert:** Double fertilization produces two distinct products: a diploid zygote (develops into embryo) and a triploid primary endosperm nucleus (develops into nutritive endosperm).

• **Alert:** Filiform apparatus is located in the synergids at the micropylar end; its essential function is to guide the entry of the pollen tube into the embryo sac.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Why is sporopollenin considered the most resistant biological material? What is the function of germ pores where sporopollenin is absent? [Score: 2 Marks]

Answer & Key Valuation Steps:

• Resistant nature: Sporopollenin can withstand high temperatures, strong acids, and strong alkalis; no known biological enzyme can degrade it, preserving pollen in fossils. **[1 Mark]**

• Function of germ pores: Apertures on exine where sporopollenin is absent, allowing the emerging pollen tube to grow out during pollen germination. **[1 Mark]**

2. Geitonogamy is functionally cross-pollination but genetically autogamous. Justify this statement. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Functionally cross-pollination: Involves the transfer of pollen grains from the anther to the stigma of another flower on the same plant, requiring external pollinating agents. **[1 Mark]**

• Genetically autogamous: Because both flowers are borne on the same parent plant, the pollen grains are genetically identical to the female parent, causing self-pollination. **[1 Mark]**

3. Describe the cellular organization of a mature, 7-celled, 8-nucleate female gametophyte (embryo sac) in angiosperms. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Egg apparatus (micropylar end): Consists of two synergids with cellular finger-like thickenings (filiform apparatus) and one central female egg cell. **[1 Mark]**

• Antipodal cells (chalazal end): Three cells located at the opposite chalazal pole, involved in vegetative nutrient uptake before degenerating. **[1 Mark]**

• Large central cell: Contains two polar nuclei situated in the center, which fuse to form a diploid secondary nucleus prior to triple fusion. **[1 Mark]**

4. Explain the events of double fertilization in angiosperms. Why is this phenomenon unique to flowering plants? [Score: 3 Marks]

Answer & Key Valuation Steps:

• Pollen tube discharge: The pollen tube releases two haploid male gametes into the cytoplasm of one of the synergids. **[0.75 Mark]**

• Syngamy: One male gamete (n) moves toward the egg cell and fuses with its nucleus, producing a diploid zygote (2n) that develops into the embryo. **[0.75 Mark]**

• Triple fusion: The second male gamete (n) fuses with the two haploid polar nuclei (2n) in the central cell to produce a triploid Primary Endosperm Nucleus (PEN, 3n). **[0.75 Mark]**

• Uniqueness: Because two types of fusions (syngamy and triple fusion) take place simultaneously in the embryo sac, it is termed double fertilization, a hallmark found solely in angiosperms. **[0.75 Mark]**

5. List and explain four outbreeding devices developed by angiosperms to prevent inbreeding depression and promote cross-pollination. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Pollen release and stigma receptivity not synchronized (Dichogamy): Anther dehisces before stigma matures (protandry) or stigma matures before pollen is shed (protogyny). **[0.75 Mark]**

• Spatial separation (Herkogamy): Anther and stigma are placed at different heights or positions so pollen cannot contact the stigma of the same flower. **[0.75 Mark]**

• Self-incompatibility: A genetically determined mechanism that inhibits self-pollen germination or pollen tube growth in the pistil. **[0.75 Mark]**

• Production of unisexual flowers (Dicliny): Monoecious plants (castor, maize) prevent autogamy but permit geitonogamy; dioecious plants (papaya) prevent both autogamy and geitonogamy. **[0.75 Mark]**

6. Differentiate between albuminous (endospermic) and non-albuminous (exalbuminous) seeds with two botanical examples for each. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Albuminous seeds: Retain a portion of endosperm as food storage tissue because it is not completely consumed during embryonic development (e.g., Castor, Maize, Wheat, Barley). **[1.5 Marks]**

• Non-albuminous seeds: Completely consume the endosperm during embryonic development, storing food reserves inside swollen fleshy cotyledons (e.g., Pea, Groundnut, Bean). **[1.5 Marks]**

7. A plant geneticist examines an angiosperm seed with a diploid maternal seed coat (2n = 24). Determine the chromosome numbers of: (a) Aleurone layer, (b) Embryo cotyledon, (c) Synergid cells, and (d) Endothecium of the parent anther. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Haploid number deduction: Since 2n = 24 in maternal tissue, the basic haploid number is n = 12. **[1 Mark]**

• (a) Aleurone layer: Part of the triploid endosperm (3n) = 3 * 12 = 36 chromosomes. **[0.75 Mark]**

• (b) Embryo cotyledon: Diploid embryonic tissue (2n) = 2 * 12 = 24 chromosomes. **[0.75 Mark]**

• (c) Synergid cells: Haploid female gametophyte cells (n) = 12 chromosomes. **[0.75 Mark]**

• (d) Endothecium: Diploid sporophytic wall layer of the parent anther (2n) = 24 chromosomes. **[0.75 Mark]**